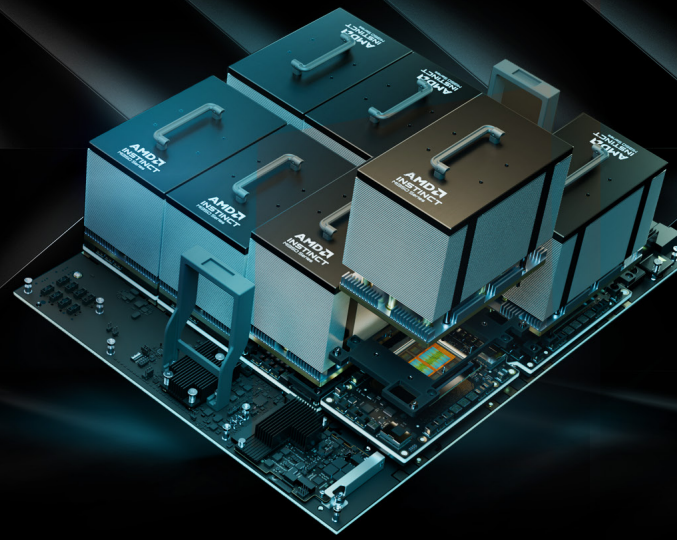


AMD INSTINCT™ MI350X PLATFORM

ADVANCED GPU SOLUTION FOR AI, INFERENCE,
AND TRAINING



ULTIMATE AI & HPC PERFORMANCE

Built on the cutting-edge 4th Gen AMD CDNA™ architecture, the AMD Instinct™ MI350 Platform features eight GPUs with powerful and energy-efficient cores, maximizing performance per watt to drive the next era of AI and HPC innovation.

With exceptional efficiency and performance, the platform excels at high-speed inference, training massive AI models, and tackling the most complex computational challenges—from scientific simulations to large-scale data processing and advanced modeling.

BREAKTHROUGH AI ACCELERATION WITH ADVANCED PRECISION & HUGE MEMORY

The AMD Instinct MI350 Platform redefines AI acceleration with next-gen MXFP6 and MXFP4 datatype support, optimizing efficiency, bandwidth, and energy use for lightning-fast, power-efficient AI inference and training.

AI PEAK THEORETICAL PERFORMANCE		W/SPARSITY
FP16 VECTOR (TFLOPS)	1,153.6	N/A
FP16 MATRIX (PFLOPS)	18.4552	36.9104
BFLOAT16 MATRIX (PFLOPS)	18.4552	36.9104
INT8 MATRIX (POPS)	36.9096	73.8192
MXFP8 (PFLOPS)	36.9096	N/A
OCP-FP8 (PFLOPS)	36.912	73.824
MXFP6 (PFLOPS)	73.816	N/A
MXFP4 (PFLOPS)	73.816	N/A
HPC PEAK THEORETICAL PERFORMANCE		
FP64 VECTOR (TFLOPS)	576.8	
FP32 VECTOR (PFLOPS)	1.2	
FP64 MATRIX (TFLOPS)	576.8	
FP32 MATRIX (PFLOPS)	1.2	
DECODERS AND VIRTUALIZATION		
DECODERS†	32 groups for HEVC/H.265, AVC/H.264, V1, or AV1	
JPEG/MJPEG CODEC	320 cores, 10 cores per group	
GPU PHYSICAL PARTITIONS	Up to 8 @ 36 GB	
MEMORY PARTITIONS	1 or 4 per module	

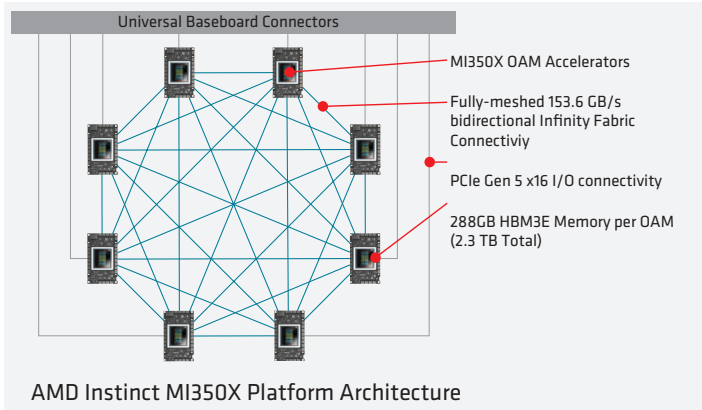
†Video codec acceleration (including at least the HEVC (H.265), H.264, VP9, and AV1 codecs) is subject to change and not operable without inclusion/installation of compatible media players. GD-176

The platform combines the power of eight GPUs on an industry-standard universal baseboard (UBB 2.0). The eight Open Compute Project (OCP) Accelerator Modules (OAMs) are connected with an AMD Infinity Fabric™ mesh that provides direct connectivity between each of the GPUs over 153.6 GB/s bidirectional links. Each MI350X connects with its peers through seven links, plus one PCIe® Gen 5 x16 connection per OAM device for upstream server and/or I/O connectivity. Each accelerator boasts 288 GB of HBM3E memory with 8 TB/s of memory bandwidth. This combines into a massive 2.3 TB of coherent, shared memory.

SEAMLESS GPU UPGRADES FOR BETTER PERFORMANCE

The platform's UBB form factor helps ensure seamless upgrades across MI300X, MI325X, and MI350X GPUs. Now you can boost performance and efficiency using your existing infrastructure, accelerating time to market while minimizing costly infrastructure changes.

SPECIFICATIONS	
FORM FACTOR	UBB 2.0 module
LITHOGRAPHY	TSMC 3nm/6nm FinFET
GPU COMPUTE UNITS	2048
MATRIX CORES	8192
STREAM PROCESSORS	131,072
PEAK ENGINE CLOCK	2200 MHz
MEMORY CAPACITY	2.3 TB HBM3E
MEMORY BANDWIDTH	8 TB/s per GPU max. peak theoretical
MEMORY INTERFACE	8192 bits
AMD INFINITY CACHE™ (LAST LEVEL)	256 MB per GPU
SCALE-UP AMD INFINITY FABRIC™ LINKS	7 (per GPU) x 153.6 GB/s
I/O TO HOST CPU	8 PCIe® Gen 5 x16 (128 GB/s)
RAS FEATURES	Full-chip ECC memory, page retirement, page avoidance
MAXIMUM TBP	1000W per module



datatypes, are designed to accelerate training while maintaining high accuracy for AI models, helping overall GPU utilization efficiency. Fast data loading and augmentation pipelines can reduce CPU-GPU bottlenecks, helping ensure high-throughput data preprocessing for large datasets.

AMD ROCm software is open source, so it is continually refined by a worldwide team of experts. Future plans include:

- Incorporating advanced memory allocation techniques to improve paging and fragmentation handling in order to support larger, more complex AI models
- Extending ROCm model and framework support
- Advancing compiler operations to help drive peak training performance
- Optimizing kernel libraries and enhancing RCCL communication for multi-GPU scaling

With latest advancements in AMD ROCm software and AMD Instinct™ MI350 Series GPUs, AMD continues deliver excellent scalability, efficiency, and innovation for next-generation AI applications.

TRUSTED BY AI LEADERS

AMD Instinct GPUs are trusted by industry leaders to drive the most demanding AI and HPC workloads with exceptional performance, reliability, and security.

PROVEN CUSTOMER SUCCESS

Major AI leaders such as Meta have chosen AMD Instinct GPUs to power large-scale AI deployments, achieving leadership price-performance for inferencing and serving as the AI infrastructure for flagship models like Llama 405B and GPT.

GROWING OEM & CLOUD ADOPTION

AMD Instinct GPUs power AI at scale, with strong adoption across leading cloud service providers (CSPs) and OEM partners. Leading and emerging CSPs are integrating AMD Instinct GPUs into their AI infrastructure for both inference and training. As demand for AI accelerates, more CSPs offer instances powered by AMD.

On the OEM front, leading manufacturers are integrating AMD Instinct Series GPUs into next-generation AI and HPC platforms. With a rapidly expanding ecosystem, AMD is delivering scalable, efficient AI solutions, empowering customers with greater choice and a competitive edge in AI innovation.

LEARN MORE

The AMD Instinct MI350X Platform is available through [AMD solution partners](#). Standard form factors facilitate adoption into enterprise servers so that you can use the same power in your data center, or in the cloud from the leading superscalars. Learn more at [AMD.com/INSTINCT](https://www.amd.com/en/instinct).

PROVEN, OPEN, LIMITLESS AI SOFTWARE

AMD ROCm software is the foundation of AMD accelerated computing, empowering AI developers to fully leverage AMD Instinct GPUs including the latest Instinct MI350 Series for both inference and training. With Day-0 support for major AI frameworks like PyTorch, TensorFlow, JAX, and ONNX Runtime, the latest advancements in ROCm further helps simplify AI model migration and deployment, optimizing hardware efficiency while minimizing code changes. Through strategic collaborations with AI leaders such as OpenAI, PyTorch, Hugging Face, Databricks and more, ROCm delivers high-performance, out-of-the-box AI solutions, enabling smooth integration, and robust partner support.



BREAKTHROUGH AI INFERENCE

The [ROCm 7.0 release](#) delivers significant advancements in AI inference, helping enable real-time processing, ultra-low latency, and increased power efficiency—critical for applications such as natural language processing and computer vision. Optimized low-latency kernels can help enhance transformer-based model execution, accelerating attention mechanisms for fast response times. With enhanced quantization support, ROCm software is designed to improve FP8, FP6, and FP4 inference, enabling efficient low-bit AI processing that reduces power consumption while maintaining accuracy. Smart memory management helps reduce inference bottlenecks, with seamless execution of large-scale AI workloads on AMD Instinct GPUs.

Improved sparse matrix computation can help boost performance in large language models (LLMs), optimizing throughput with less computational overhead. These advancements, combined with access to bi-weekly optimized containers via the AMD Infinity Hub, streamline the deployment of high-performance inference models, improving scalability and offering low operational costs.

SCALABLE AI TRAINING

The ROCm 7.0 release is designed to help enhance AI training performance and scalability, with fast convergence and optimized compute for large-scale AI models. Improved mixed-precision training, with optimized support for FP16, MXFP8, OCP-FP8, BF16, MXFP6, MXFP4

Footnote explanations are available at: <https://www.amd.com/en/legal/claims/instinct.htm>

1. MI350X Platform - PID#253461432A

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